ROLLNO:2520030609
NAME: V.Siri Chandana
SECTION:2

Skill-1

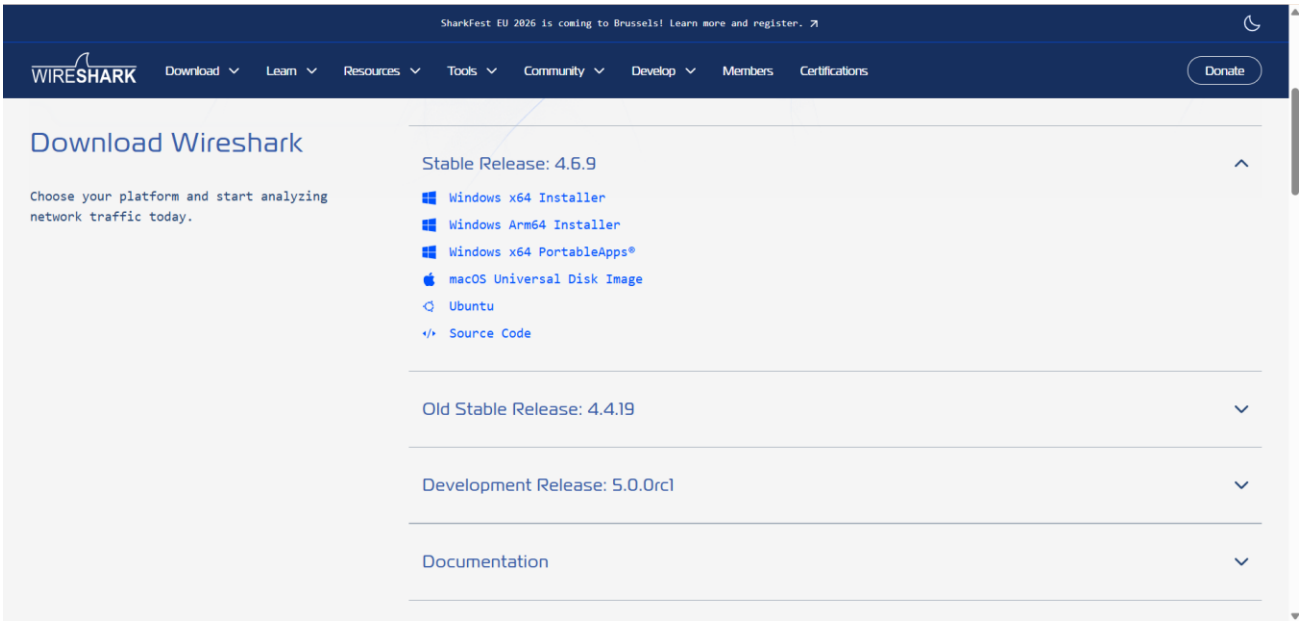
Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.

Step 1: Install Wireshark

1. Open the browser.
2. Install Wireshark [Wireshark • Go Deep | Download]
3. Install it using the default settings.
4. Open Wireshark.

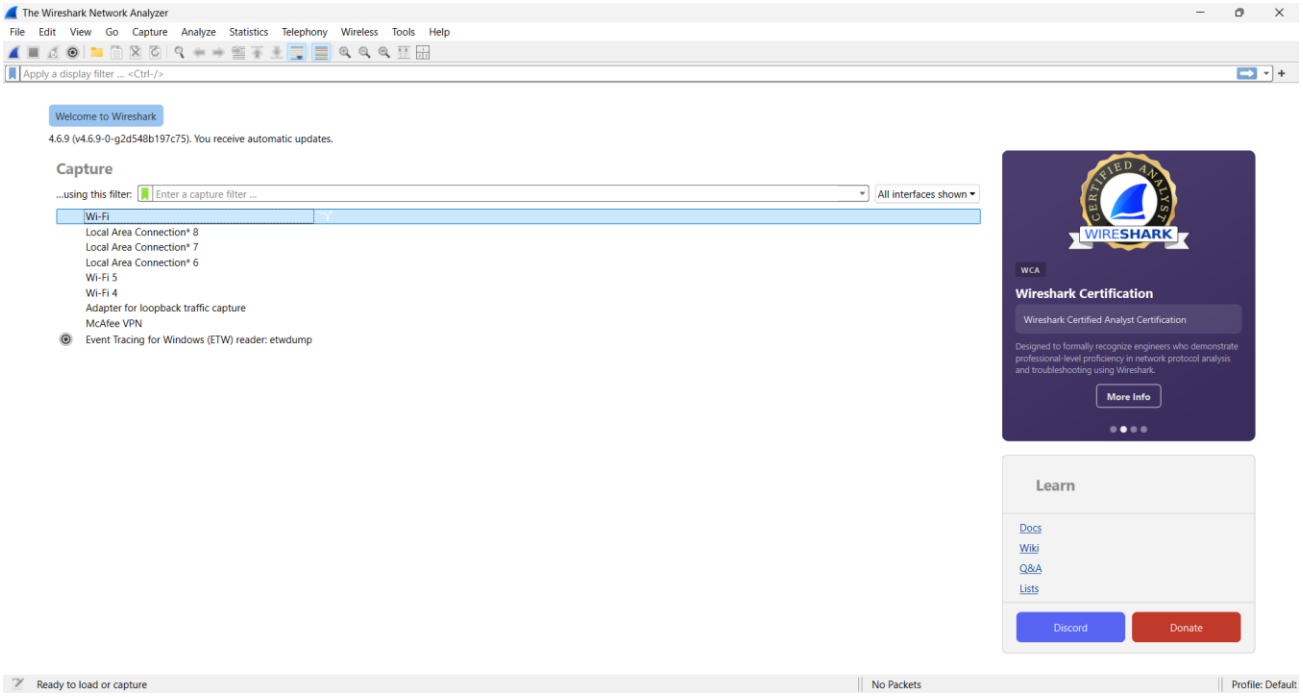


Step 2: Select your network interface

When Wireshark opens, you will see interfaces such as:

Wi-Fi, Ethernet, Bluetooth

1. If you are connected through Wi-Fi, select Wi-Fi.
2. Double-click the Wi-Fi interface.
3. Packet capture will start.

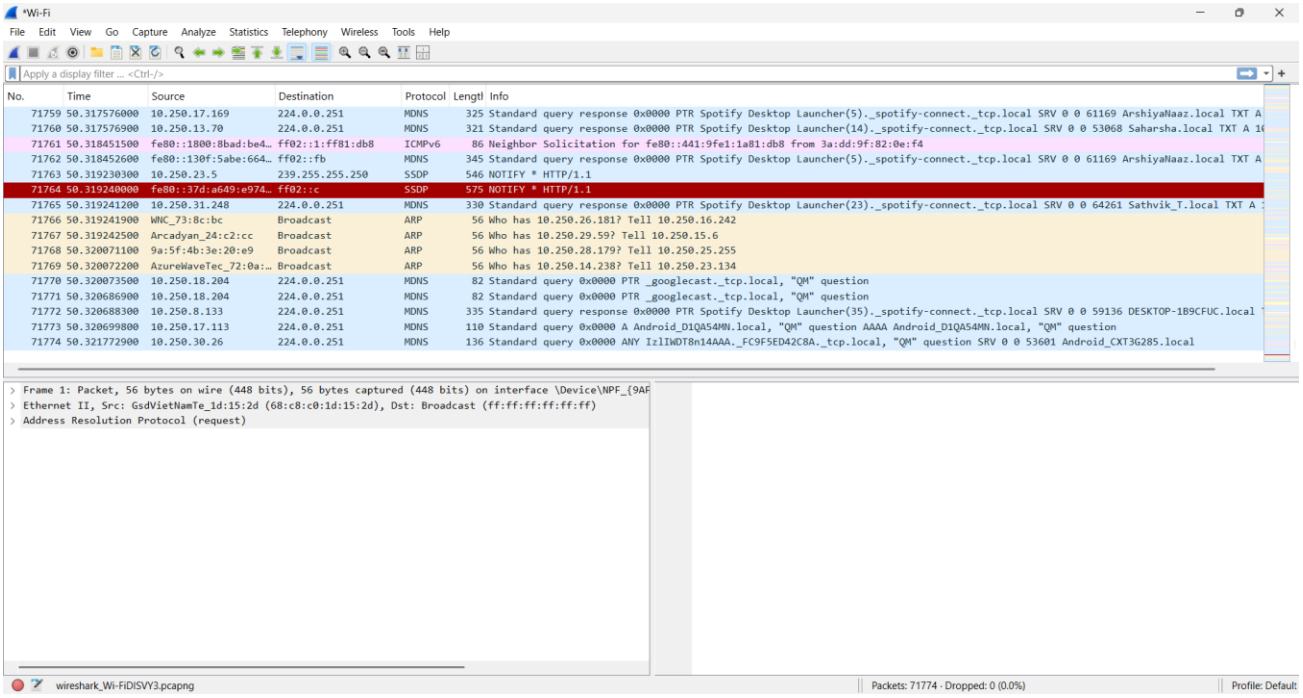


Step 3: Generate network traffic

While Wireshark is capturing:

1. Open a new browser tab.
2. Visit a website such as https://www.google.com.
3. Open two or three different websites.
4. Wait for approximately 30 seconds.
5. Return to Wireshark.
6. Click the red Stop button.

You should now see captured packets.



Step 4: Apply display filters

In the Wireshark filter bar, enter each filter separately and press Enter.

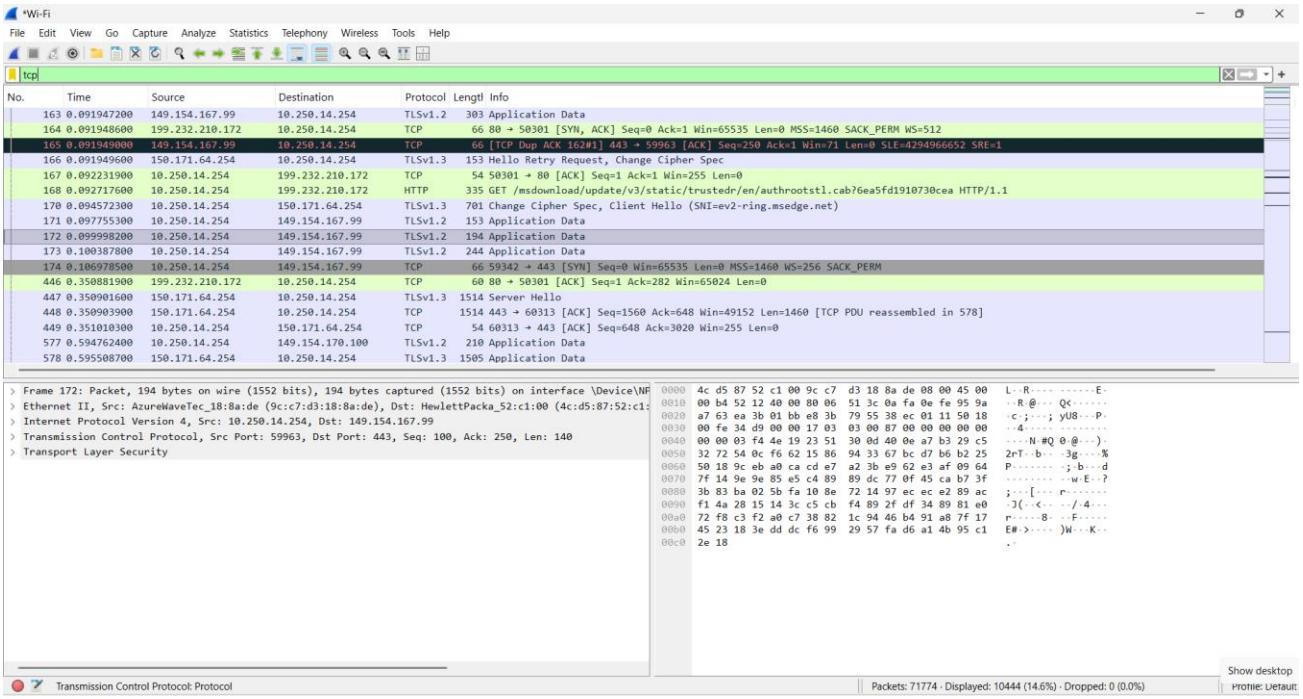
Filter	**Purpose**
tcp	Displays TCP packets
udp	Displays UDP packets
icmp	Displays ICMP packets
tcp.len > 0	Displays TCP packets carrying data

First, enter: TCP

Observe the packets displayed.

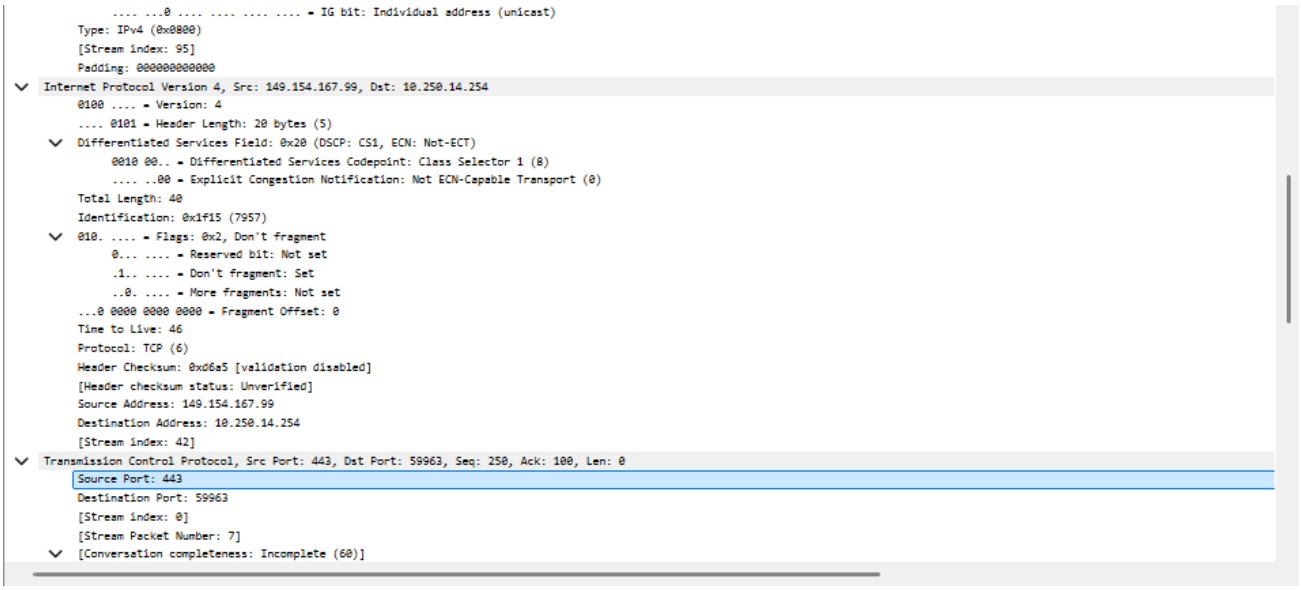
Then enter: UDP

Observe the UDP packets.



Step 5: Inspect packet headers
Select any TCP packet from the packet list.

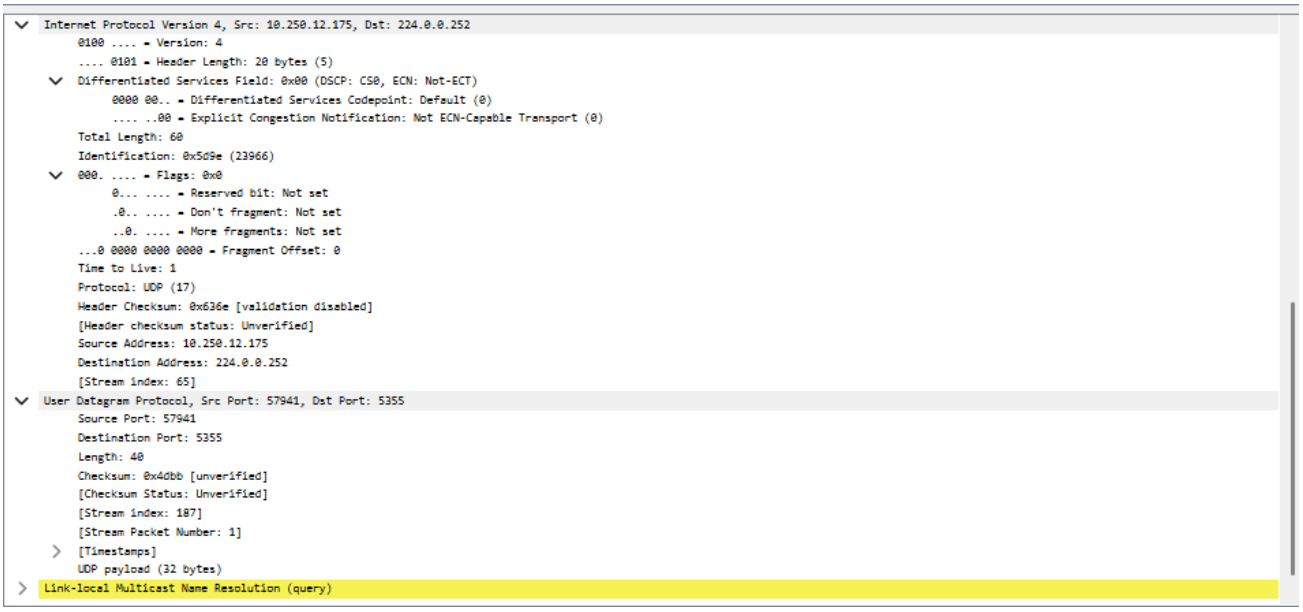
In the packet details pane, expand these sections:
1. Frame
2. Ethernet II
3. Internet Protocol Version 4
4. Transmission Control Protocol



Record the following values:

Parameter	Where to find it
Frame length	Frame
Ethernet Header	Ethernet II
Source MAC Address	Ethernet II
Destination MAC Address	Ethernet II
IPv4 Header Length	Internet Protocol Version 4
IPv4 Total Length	Internet Protocol Version 4
TCP Header Length	Transmission Control Protocol
TCP Segment Length	Transmission Control Protocol

Repeat the process for one UDP packet. Observe the UDP header and its length field.



Step 6: Calculate header overhead

For a basic Ethernet + IPv4 + TCP packet without TCP options:

- Ethernet header = 14 bytes
- IPv4 header = 20 bytes
- TCP header = 20 bytes
- Application data = 1460 bytes
- FCS = 4 bytes, if included in the frame representation

Total size including FCS:

$$14 + 20 + 20 + 1460 + 4 = 1518 bytes$$

Total overhead:

$$14 + 20 + 20 + 4 = 58 bytes$$

Payload efficiency:

$$\frac{1460}{1518} \times 100 = \boxed{96.18\%}$$

Note: Wireshark may not capture or display the Ethernet FCS. Use the actual captured packet fields when recording your observations.

Step 7:
Fill in your observation table

Record values from your own Wireshark capture.

Parameter	**TCP Packet**	**UDP Packet**
Frame Length	1514 bytes	74 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	1500 bytes	60 bytes
Transport Header Length	20 bytes	8 bytes
Payload Length	1460 bytes	32 bytes

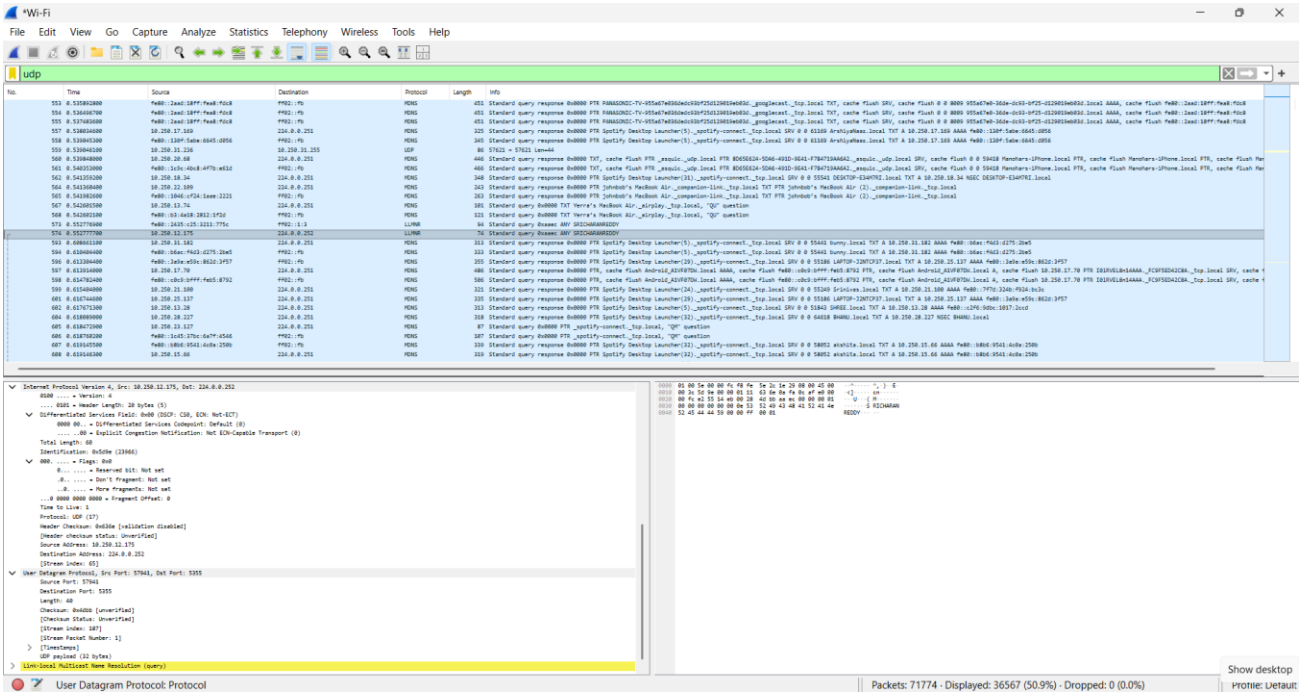
Steps to save a capture file

Step 1: Start capturing packets

1. Open Wireshark.
2. Select your Wi-Fi or Ethernet interface.
3. Double-click the interface to start capturing packets.

Step 2: Stop the capture

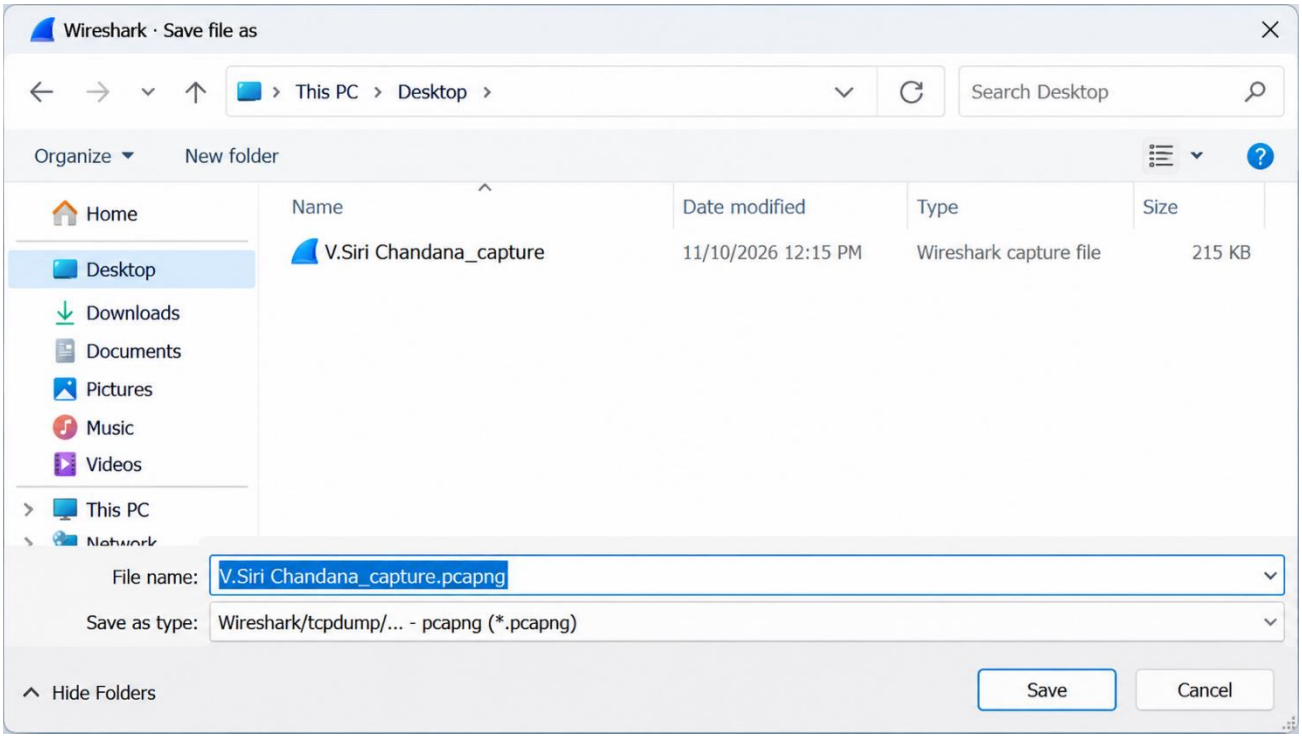
Once you have generated sufficient network traffic:
1. Click the red Stop button (top-left corner).
2. Your captured packets will remain visible on the screen.



Step 3: Save the capture

1. Click File in the menu.
2. Select Save As.
3. Choose the folder where you want to save the file.
4. Enter a filename.
5. Keep the file format as pcapng.
6. Click Save.

For example: sender_capture.pcapng



How to open a saved capture file

1. Open Wireshark.

2. Click File → Open.

3. Select your .pcapng or .pcap file.

4. Click Open.

5. Enter icmp in the display filter bar to view ICMP packets.

Answer these questions

1. What is encapsulation?

Answer:

Encapsulation is the process of adding headers to data at each network layer before sending it from one device to another.

For example:

- Transport layer adds a TCP or UDP header.
- Network layer adds an IP header.
- Data Link layer adds an Ethernet header.

2. What is the purpose of the Ethernet header?

Answer:

The Ethernet header is used to deliver data between devices in a local network.

It contains:

- Source MAC address.
- Destination MAC address.
- Type of protocol.

3. What information is stored in the IP header?

Answer:

The IP header contains information required to send packets from the source device to the destination device.

It stores:

- Source IP address.
- Destination IP address.
- IP version.
- Packet length.
- Time to Live (TTL).
- Protocol information (TCP or UDP).

4. What is the difference between TCP and UDP headers?

Answer:

TCP and UDP are transport layer protocols used for communication between devices.

TCP Header	UDP Header
Minimum 20 bytes	8 bytes
Provides reliable communication	Does not guarantee delivery
Uses acknowledgements	Does not use acknowledgements
Connection-oriented	Connectionless
More overhead	Less overhead

5. Why does each protocol add a header?

Answer:

Each protocol adds a header to provide necessary information for communication.

Headers help identify the source and destination, deliver data correctly, control communication and detect errors.

6. What is packet overhead?

Answer:

Packet overhead is the extra data added to a packet by different network protocols.

It includes Ethernet, IP and TCP or UDP headers. This information is required for proper network communication.

7. How does header overhead affect network efficiency?

Answer:

Header overhead reduces network efficiency because some part of the transmitted packet is used for headers instead of actual application data.

When header size increases, less space is available for useful data. Therefore, network efficiency decreases.

Formula:

$$\text{Payload Efficiency} = \frac{\text{Application Data}}{\text{Total Packet Size}} \times 100$$

Your TCP calculation:

$$\frac{1460}{1518} \times 100 = 96.18\%$$